Protocolos de comunicaciones de alta velocidad

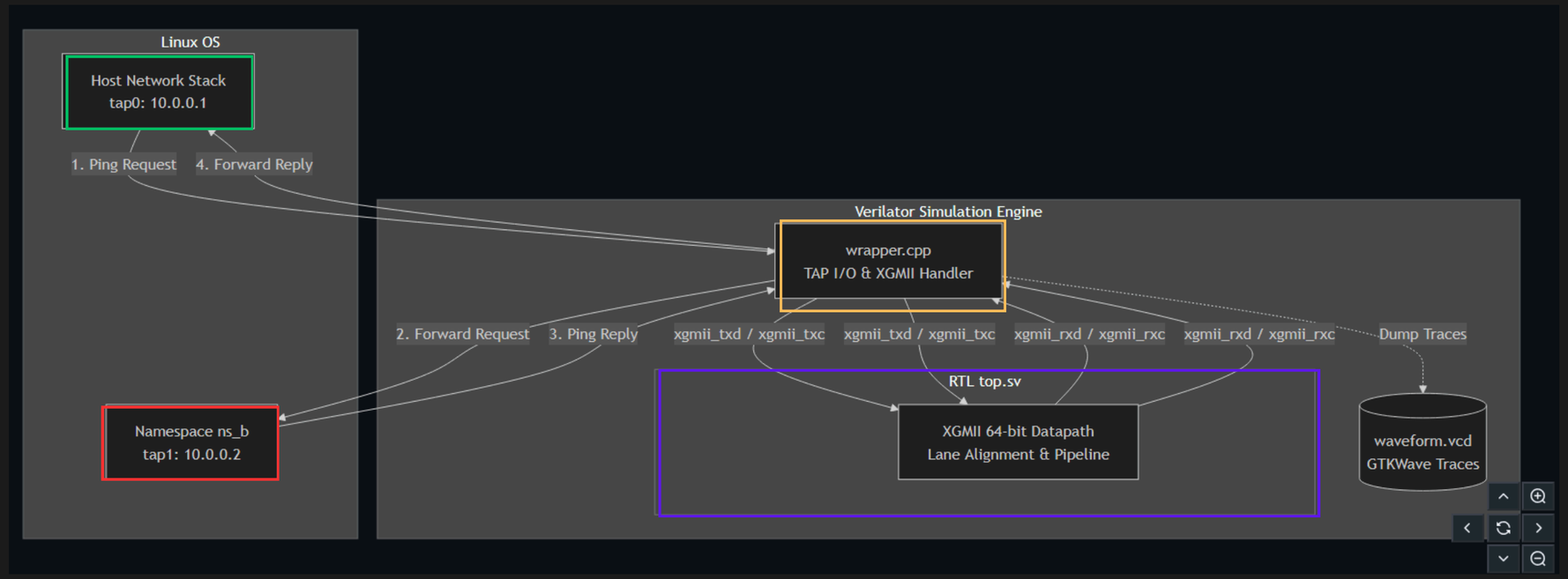
Ethernet-phy

Lab2 XGMII Interface & Reconciliation Sublayer (RS)

Fundación Fulgor

Facundo Guerrero

Architecture Setup



- Consists of two virtual interfaces, Tap0 and Tap1, which can communicate with each other (for example, using the *ping* command).
- In between them, there is a *wrapper* that takes the data and processes it according to the specific case being studied on each laboratory. Then injects the data into the input pins of the *top.sv* circuit. At the same time, it records a *.vcd* file for waveform visualization.
- The digital circuit *top.sv* is a module used to simulate the hardware. It assigns the signals and, when necessary, creates debug signals.
- tcpdump is used in parallel to generate a *.pcap* file for subsequent analysis.
- In this case, Lab 2, the data (TXD) will be encoded, and there will be an another signal (TXC) that indicates control fields
- Finally, the packets are analyzed using Wireshark and GTKWave.

Architecture Setup

```
~/Documentos/fulgor/labs/ethernet-phy-labs/scripts main
❯ sudo ./setup_netns.sh
[sudo] contraseña para facusu:
═══ 1. Limpiando configuración previa ═══
═══ 2. Creando interfaces TAP y Namespace de red ═══
═══ 3. Configurando Direcciones IP y Estado ═══
═══ 4. Deshabilitando Offloading para Simulación RTL ═══
═══ 5. Verificando estado del entorno ═══
→ Interfaz tap0 (Host):
5: tap0: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 5a:f4:04:87:86:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/24 scope global tap0
       valid_lft forever preferred_lft forever
→ Interfaz tap1 (Namespace ns_b):
6: tap1: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether ce:e0:bc:43:fd:85 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 scope global tap1
       valid_lft forever preferred_lft forever
================================================
Entorno preparado con éxito. Listo para la simulación.
================================================

~/Documentos/fulgor/labs/ethernet-phy-labs/scripts main
❯
```

1. We run *setup_netns.sh* to create the TAPS

```
~/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet main
❯ verilator --cc --trace --exe --build -j 0 -Wall -Wno-DECLFILENAME wrapper.cpp top.sv -o emulator

make: se entra en el directorio '/home/facusu/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet/obj_dir'
g++  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DVM_TR
DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-linka
 -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -Os  -c -o wrapper.o ../wrapper.
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated.o /usr/shar
d.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated_vcd_c.o /us
rilated_vcd_c.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated_threads.o /
verilated_threads.cpp
python3 /usr/share/verilator/bin/verilator_includer -DVL_INCLUDE_OPT=include Vtop.cpp Vtop___024root__0.cpp Vtop__Trace__0.cpp Vtop___024root__Slow.cpp Vtop__
Syms__Slow.cpp Vtop__Trace__0__Slow.cpp Vtop__TraceDecls__0__Slow.cpp > Vtop__ALL.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o Vtop__ALL.o Vtop__ALL
g++    wrapper.o verilated.o verilated_vcd_c.o verilated_threads.o Vtop__ALL.a    -pthread -lpthread -latomic   -o emulator
rm Vtop__ALL.verilator_deplist.tmp
make: se sale del directorio '/home/facusu/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet/obj_dir'
- V e r i l a t i o n   R e p o r t: Verilator 5.050 2026-07-01 rev v5.050
- Verilator: Built from 0.027 MB sources in 2 modules, into 0.031 MB in 8 C++ files needing 0.000 MB
- Verilator: Walltime 6.998 s (elab=0.014, cvt=0.027, bld=6.950); cpu 0.018 s on 4 threads; allocated 346.449 MB

~/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet main
❯ sudo ./obj_dir/emulator
[TAP] Interfaz tap0 vinculada correctamente (host).
[TAP] Interfaz tap1 vinculada correctamente (ns_b).
[EMULADOR LAB 1] Ejecutando simulación...
```

2. We use verilator to compile and generate an executable taking wrapper and top.sv files. This create an emulation executable

```
~/Documentos/fulgor/labs/ethernet-phy-labs/lab2-xgmii-64bit main
❯ ip addr show dev tap0
5: tap0: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 5a:f4:04:87:86:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/24 scope global tap0
       valid_lft forever preferred_lft forever

~/Documentos/fulgor/labs/ethernet-phy-labs/lab2-xgmii-64bit main
❯ sudo ip netns exec ns_b ip addr show dev tap1
6: tap1: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether ee:40:c4:c6:cf:5c brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 scope global tap1
       valid_lft forever preferred_lft forever

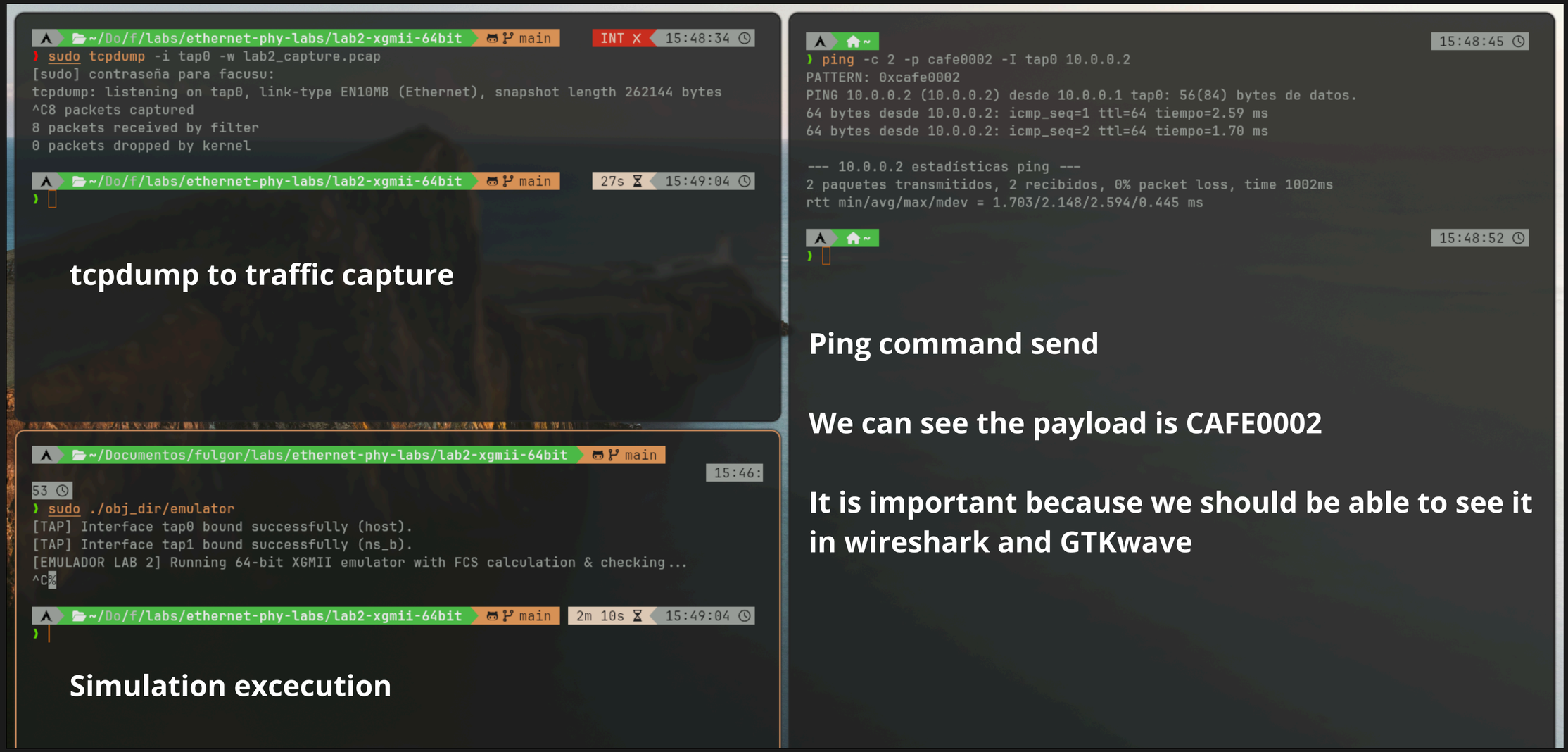
~/Documentos/fulgor/labs/ethernet-phy-labs/lab2-xgmii-64bit main
❯
```

3. Finally there are the two taps and its two MACs addresses and ips

TAP	IP	MAC
Tap0	**10.0.0.1**	**5a:f4:04:87:86:e7**
Tap1	**10.0.0.2**	**ee:40:c4:c6:cf:5c**

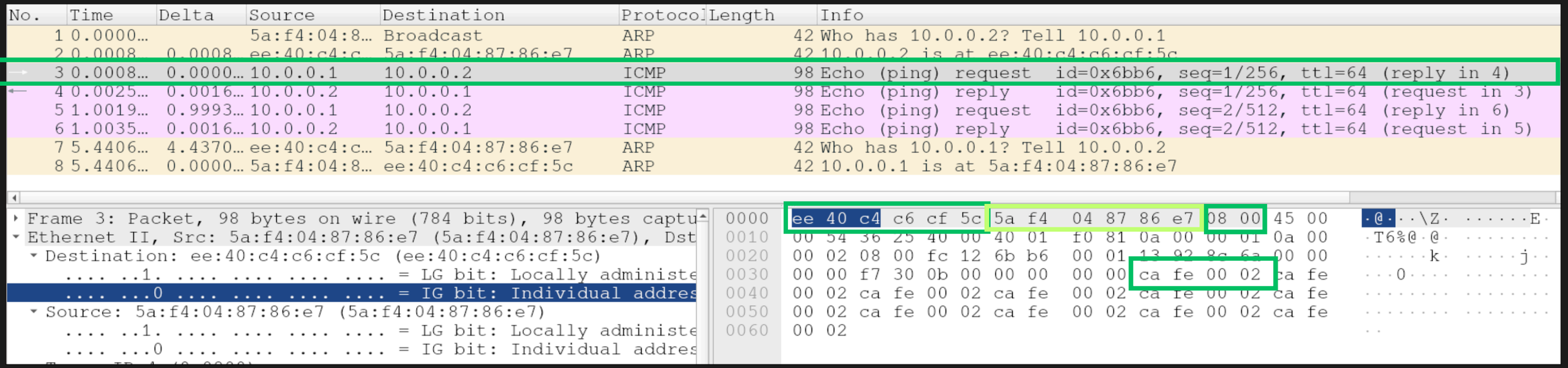
Execution

It is executed simultaneously in three terminals.



Analysis

Now we analyze the packet of interest, the Ping Request from Tap0 to Tap1.

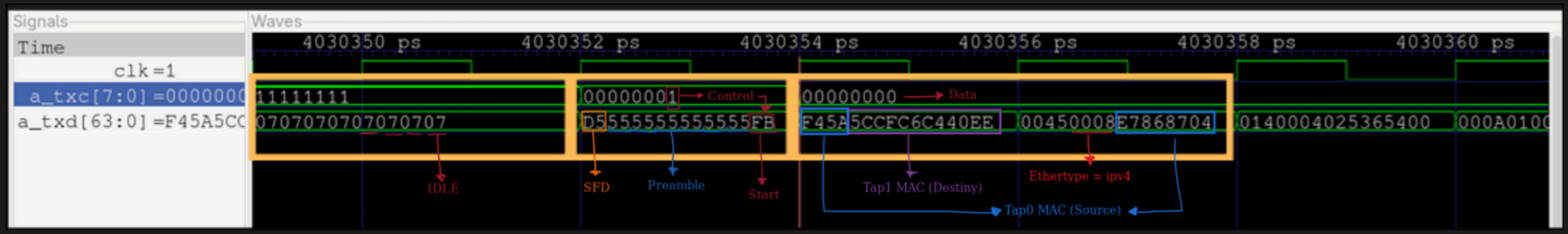


The packet starts with tap1 MAC address = ee:40:c4:c6:cf:5c because is the destination.

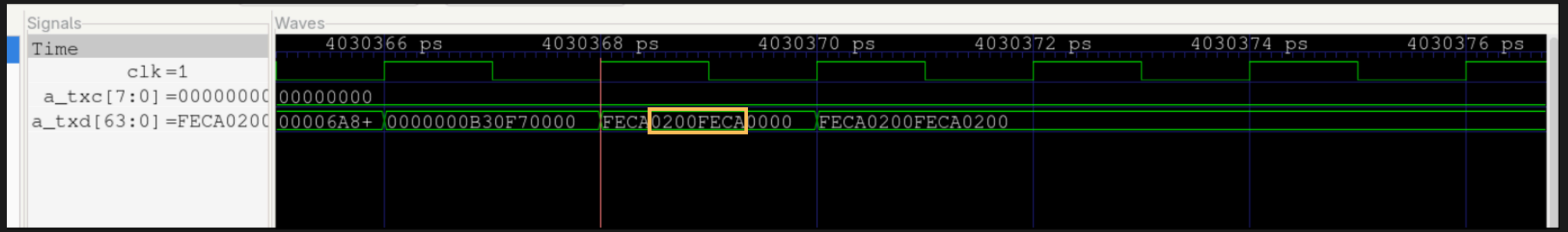
Then MAC source (tap0 MAC = 5a:f4:04:87:86:e7) and ethertype = 0800

And we can see the payload CAFE0002 that we defined previously

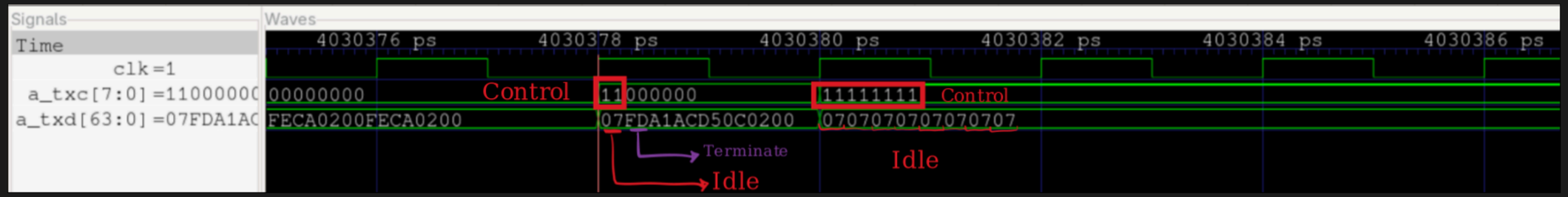
Analysis



We can see how it comes from the idle state, and then move on to the first fields, such as the start preamble, SFD, and MAC address.



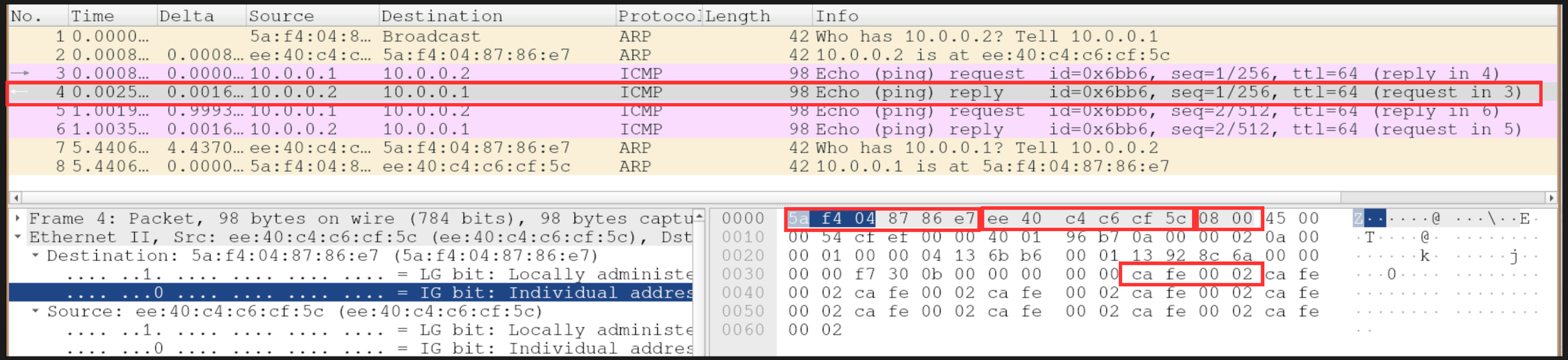
CAFE0002 Payload



Finally the last fields like Terminate with TXC bits in 1 and idle again between this packet finish and the beginning of the next one

Analysis

Now we analyze the Ping reply from tap1 to tap0

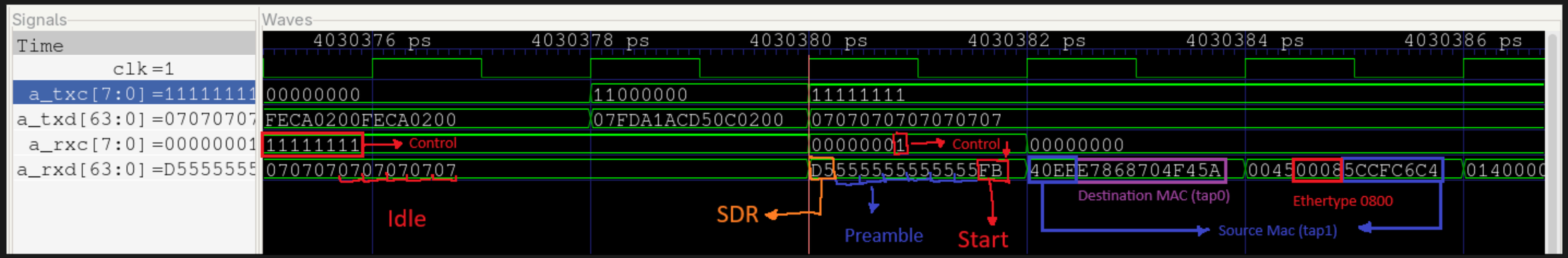


The packet starts with tap0 MAC address = 5a:f4:04:87:86:e7 because is the destination.

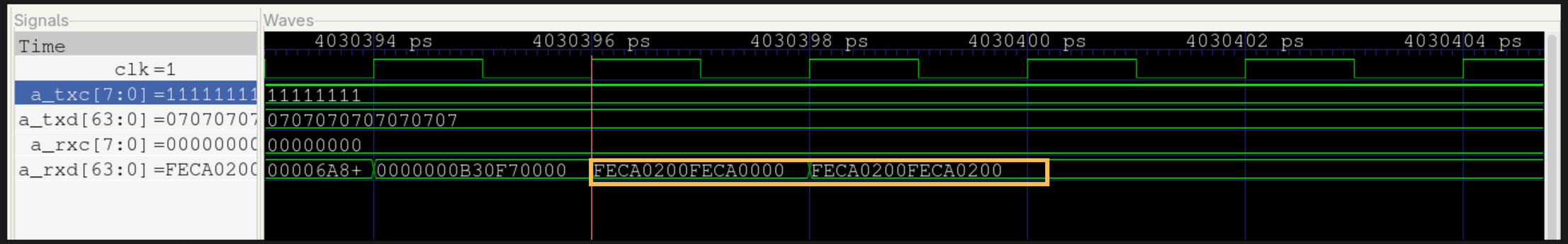
Then MAC source (tap0 MAC = ee:40:c4:c6:cf:5c) and ethertype = 0800

And we can see the payload CAFE0002 that we defined previously

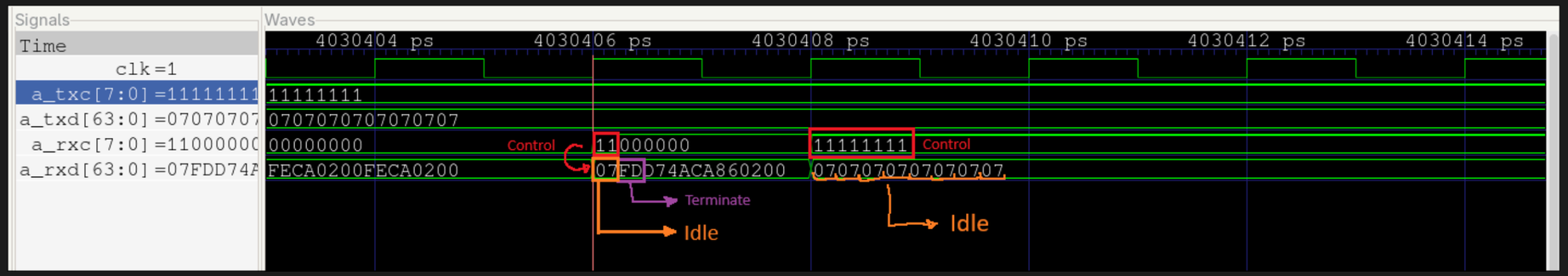
Analysis



We can see how it comes from the idle state, and then move on to the first fields, such as the start preamble, SFD, and MAC address.



CAFE0002 Payload



Finally the last fields like Terminate with TXC bits in 1 and idle again between this packet finish and the beginning of the next one